

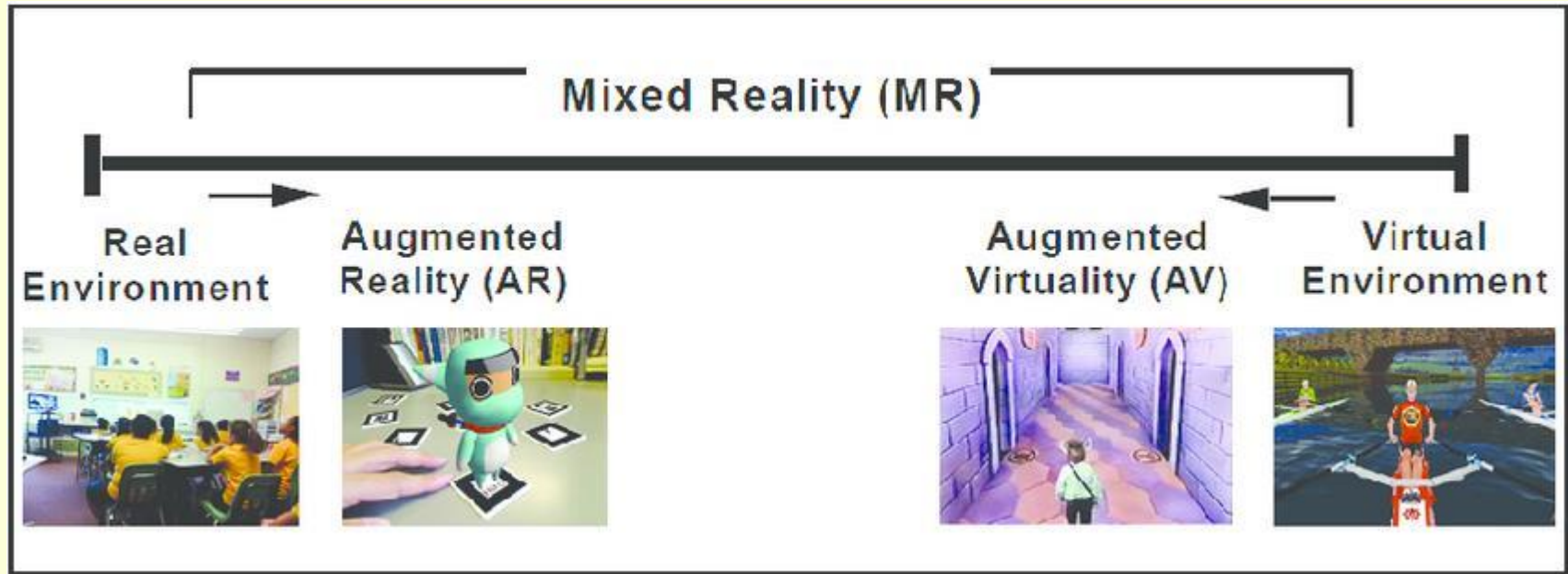
Chapter Five:

Augmented, Virtual, and Mixed Reality (Immersive Tech)

Introduction

- The technologies which allow the user to engage naturally with the blended reality by the integration of virtual content with the physical environment are known as ***Immersive technologies***.
- Such kinds of technologies create distinct experiences by merging the physical world with digital or simulated reality.
- ***Augmented reality (AR)*** and ***virtual reality (VR)*** are two principal types of immersive technologies.
- These technologies share many of the same qualities.
- However, **AR** blends computer-generated information onto the user's real environment, while **VR** uses computer-generated information to provide a full sense of immersion.

Overview of AR, VR, and MR



Overview of **Augmented Reality (AR)**

- The fundamental idea of **Augmented reality (AR)** is:
 - ✓ To **combine**, or **mix**, the **view of the real environment** with additional, **virtual content** that is presented through *computer graphics*.
- **Augmented reality (AR)** allows users to **overlay** computer- generated content in the real world.
- **Augmented reality** is the **integration of digital information** with the **user's environment** in real-time.

- In **Augmented Reality**, users see and interact with the real world while digital content is added to it.



- **By using AR**, the information about the surrounding **real world** becomes **interactive** and **digitally usable**.
- Through **augmented vision**, a user can **digitally interact** with and adjust information about their surrounding environment.



Overview of Virtual Reality (VR)

- **Virtual Reality (VR)** is **fully immersive**, which tricks your senses into thinking you're in a different environment or world apart from the real world
- Using a **Head-Mounted Display (HMD)** or **headset**, you'll **experience** a **computer-generated world** of imagery and sounds in which you can **manipulate objects** and **move around** using haptic controllers while tethered to a console or PC.
- Also called **computer-simulated reality**.



Overview of Mixed Reality (MR)

- **Mixed Reality (MR)** is the **merging** of **real** and **virtual worlds** to produce new environments and visualizations where **physical** and **digital objects co-exist** and **interact in real-time**.
- Also referred as **hybrid reality**.



Example: https://www.youtube.com/watch?v=e-n90xrVXh8&ab_channel=CNET

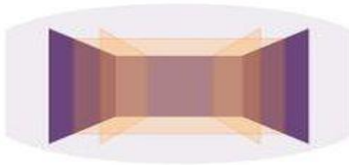
Overview of Mixed Reality (MR)



Differences Between VR, AR, MR

VIRTUAL REALITY (VR)

Fully artificial environment



Full immersion in virtual environment

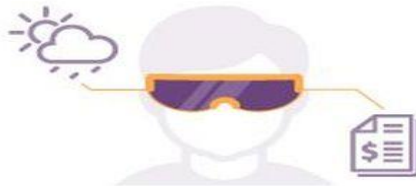


AUGMENTED REALITY (AR)

Virtual objects overlaid on real-world environment



The real world enhanced with digital objects

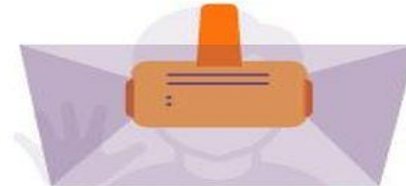


MIXED REALITY (MR)

Virtual environment combined with real world



Interact with both the real world and the virtual environment



The Architecture of AR Systems

- The first Augmented Reality Systems (ARS) were usually designed with a basis on three main blocks:
 - ✓ Infrastructure Tracker Unit,
 - ✓ Processing Unit, and
 - ✓ Visual Unit

- The **Infrastructure Tracker Unit** was responsible for collecting data from the real world, sending them to the Processing Unit.
- **Processing Unit** is going to mix the virtual content with the real content and sent the result to the Video Out module of the Visual Unit.
- Some designs used a Video In, to acquire required data for the Infrastructure Tracker Unit.

- The **Visual Unit** can be classified into two types of system, depending on the followed visualization technology:
 - ✓ **Video see-through:** uses a Head-Mounted Display (HMD) that employs a video-mixing and displays the merged images on a closed-view HMD.
 - ✓ **Optical see-through:** uses an HMD that employs optical combiners to merge the images within an open-view HMD.

Head-Mounted Display (HMD)

- **HMD** is a display device (AR equipment) worn on the head or as part of a helmet; that has a small display optic in front of one (monocular HMD) or each eye (binocular HMD).
- A typical HMD has one or two small displays, with lenses and semi-transparent mirrors embedded in eyeglasses (also termed data glasses), a visor, or a helmet.
- HMDs differ in whether they can display only **computer-generated imagery (CGI)**, or only live imagery from the physical world, or combination.

- Most HMDs can display only a computer-generated image, sometimes referred to as virtual image.
- Some HMDs can allow a CGI to be superimposed on real-world view, and this is sometimes referred to as augmented reality (AR) or mixed reality (MR).
- Combining real-world view with CGI can be done by projecting the CGI through a partially reflective mirror and viewing the real world directly. This method is often called *optical see-through*.

- Combining real-world view with CGI can also be done electronically by accepting video from a camera and mixing it electronically with CGI.
- An HMD has many uses including gaming, aviation, engineering, and medicine.

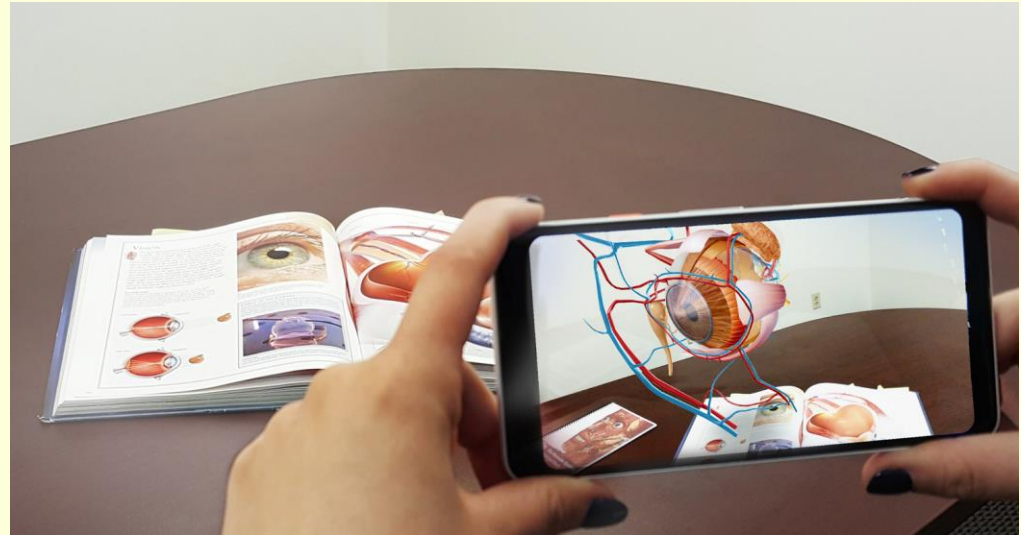


- HMDs lack in several aspects, such as ergonomics, high prices and relatively low mobility due to their sizes and connectivity features.
- An additional problem involving HMD is the interaction with the real environment, which places virtual interactive zones to the user, making the collision with these zones hard due to the difficulty to interact with multiple points in different depths.

- Alternative approaches to developing ARS involve the use of **monitors** and **tablets**.
- **Monitors** are used as an option for indirect view since the user doesn't look directly into the mixed world.
- **Tablets** are used in direct view since the user points the camera to the scene and looks directly into the mixed world.
- Both approaches still have difficulties in getting a collision.

Application of AR in Education

- Higher engagement, Higher retention, Boost intellectual curiosity, Interactive lessons, Affordable learning materials *etc* ..



- **Affordable learning materials:**

- ✓ Using AR technology allows for avoiding investments in physical materials.
- ✓ Students can get access to learning materials and interact with them anytime



- **Interactive lessons:**
 - ✓ Students can view models on their own smartphones and get a better idea of the concepts they are studying.
 - ✓ Increases engagements and reinforces the learning.



- **Higher engagement:**
 - ✓ When teachers integrate augmented reality into their lectures, they attract the attention of their students and make lessons more effective.



- **Higher retention:**

- ✓ Using the AR app, students can get access to augmented models that represent any real objects from a famous monument or work of art to a molecule.



- **Boost intellectual curiosity:**
 - ✓ AR makes students more excited about learning certain subjects.
 - ✓ AR help them learn new ideas and develop their critical thinking skills.



Application of AR in Medicine

- Education of future doctors, surgery (minimally invasive surgery), diagnostics, detect mental illnesses *etc* ..



- **Describing symptoms:**
 - ✓ AR helps patients to understand their conditions and describe correctly their symptoms.

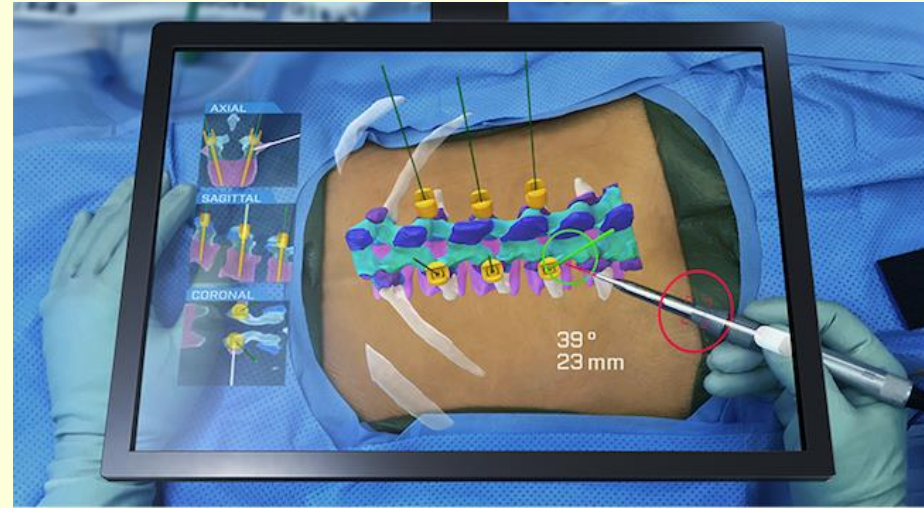


- **Nursing care:**
 - ✓ AR enabled handheld scanner projects on the skin and shows the patients' veins.



- **Surgery:**

- ✓ Three-dimensional reconstructions of organs or tumors will help surgeons become more efficient at surgery operations.



- **Ultrasounds:**

- ✓ AR software companies are developing handy ultrasound scanner, which with the help of smart glasses works as a traditional ultrasound scanner.



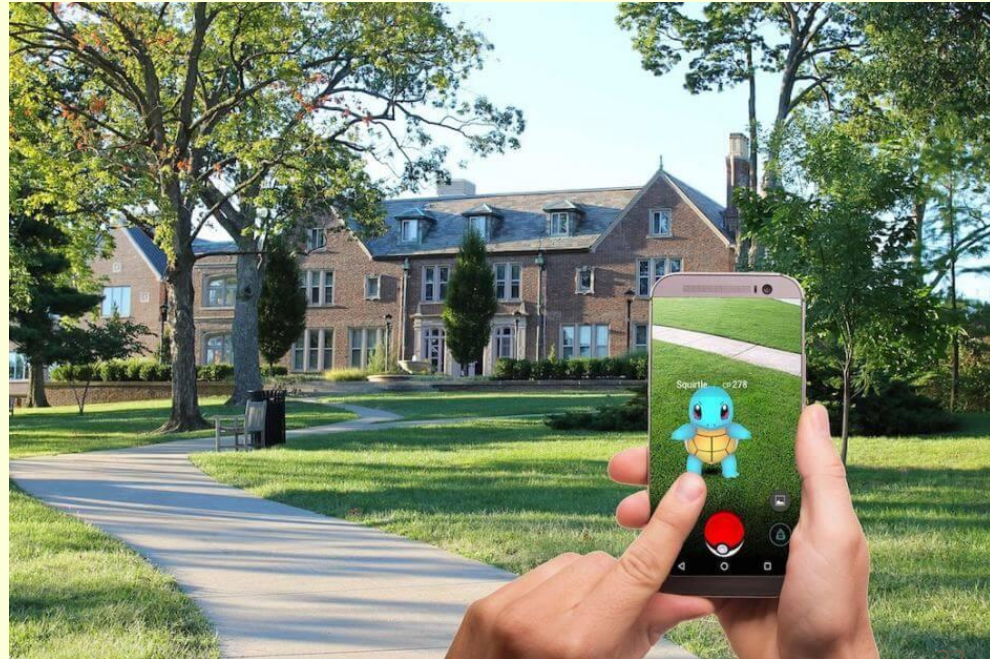
- **Navigation:**

- ✓ AR can be useful to provide information about the nearest hospitals.

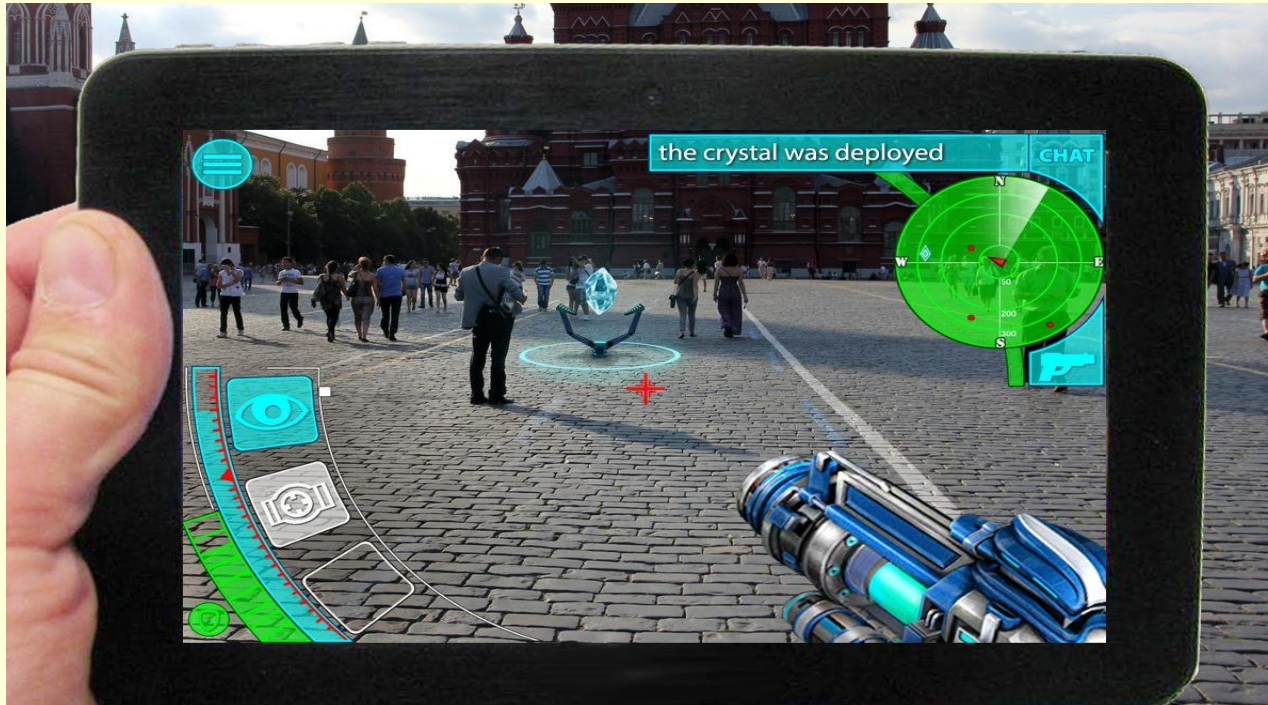


Application of AR in Entertainment

- AR in games, AR in music, AR on TV, AR in eSports, AR in the theater.



- **AR in games:**
 - ✓ AR games were praised for increasing physical activity in people.



- **AR in music:**

- ✓ Augmented reality can enhance live performances by illustrating the story told by a track or displaying the way it was created by the band.
- ✓ Augmented reality can provide complete information on the track or its performer.



- **AR** on **TV**:

- ✓ One way of integrating augmented reality in television is adding supplementary information to what is going on the TV screen, such as match scores, betting options, and the like.



- **AR in eSports:**
 - ✓ Augmented reality turns eSports shows into interactive experiences allowing the watchers to become participants.



- **AR** in the **theater**:
 - ✓ The possibility to overlay virtual objects over the real environment can be used, *for example*, for subtitling in various theater shows.



More Videos

:

https://www.youtube.com/watch?v=XLP4YTpUpBI&t=2s&ab_channel=Simplearn

https://www.youtube.com/watch?v=ZFTgGi06vbM&t=231s&ab_channel=edureka%21

<https://www.youtube.com/watch?v=f9MwaH6oGEY>

https://www.youtube.com/watch?v=IEzaFR_X35U

<https://www.youtube.com/watch?v=2zsIXVqxbuc>

<https://www.youtube.com/watch?v=AJJOWemfOYI>